Cloud Computing

Cloud computing provides computing resources such as servers, storage, databases, and software over the internet.

Infrastructure as a Service (IaaS) provides virtualized computing resources. Platform as a Service (PaaS) provides an environment for application development and deployment. Software as a Service (SaaS) provides ready-to-use software over the internet.

Public cloud services are provided over shared infrastructure. Private clouds are dedicated to a single organization. Hybrid cloud combines public and private cloud environments.

Scalability allows computing resources to increase or decrease based on demand. Elasticity automatically adjusts resources according to changing workloads.

Cloud computing can reduce infrastructure management effort and support flexible resource usage.